Exact Compression and Finite Jet Certificates for Higher-Dimensional Seshadri Problems

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M097

October 8, 2026

Abstract

We examine the finite algebraic and analytic interfaces in the October 5, 2026 manuscript *Maximal Multipoint Seshadri Constants in Higher Dimensions*. Its claimed eventual maximality theorem concerns ordinary curve-defined constants on a fixed smooth complex polarized variety, in dimension at least three, at every sufficiently large integer point count. We do not certify that global theorem. We prove an exact cardinality formula for Laurent-row compression, construct rational normalized lifts with explicit analytic error bounds, and retain multiplication across degrees independently of the transfer parameters. We give an anisotropic missing-exponent certificate and a dimension-minimal Hermite staircase for unequal jet orders, including its exact determinant. Their combination yields finite lower-bound certificates and, given quantitative deficits and the source's geometric transfers, a degree-dependent approach to the volume scale. These are source-specific quantitative deductions from classical Vandermonde, Hermite, and sumset arguments, not a new proof of Nagata–Biran–Szemberg maximality. Contribution assessment: C3/5, because the explicit compression lifts and anisotropic deficit-to-jet certificates give notable local quantitative improvements while the global geometric construction remains an input.

1 Problem, scope, and dependency map

Let X be a smooth integral complex projective variety, $\dim X = n \geq 3$, and let L be ample with $V = L^n$. For distinct points $\mathbf{p}$ define

$$\varepsilon(X, L; \mathbf{p}) = \inf_C \frac{L \cdot C}{\sum_i \operatorname{mult}_{p_i} C},$$

where C is integral and meets at least one marked point. This is the ordinary, not a higher-cycle, constant. The source [1] asserts that there exists $r_0(X, L)$ such that for each integer $r \geq r_0$ it equals $w = (V/r)^{1/n}$ off a countable union of proper closed subsets of the configuration space $U_r(X)$. This is stronger than asymptotic maximality as $r \to \infty$, and says nothing of this form for every special tuple.

The actual higher-dimensional source, not its bundled surface companion, was read in full: its six scientific section files have 1,274 lines. Main file, README, bibliography, and figure definitions add 326 lines. Literal intervals and hashes are in the accompanying read ledger. All eleven bundled

companion texts remain unread. Reading every line of this manuscript is not an audit of every cited theorem, nor a whole-corpus proof audit.

Here is the source's dependency chain. The locators refer to its section files, with the `.tex` suffix suppressed.

Source stage	Exact role	Boundary in this report
`transfers`, 1–270	Least terms, labeled products, positive weighted limits, finite jet rank	Exact row compression and a quantitative instance of the analytic transfer are proved below.
`flag`, 24–139	Complete-intersection flag; supports of size $h^0(X, kL)$ in a volume-correct simplex	Source geometric construction; Bertini and Hilbert polynomial inputs are retained.
`reshaping`, 16–288	Node $y^2-x^2(1+x)$, real intercepts, primitive direction, repeated compressions	Read and mapped, not replaced by a finite matrix certificate.
`saturation`, 17–166	Two-degree decomposition and elongated-simplex Hermite interpolation	Quantified and extended to unequal jet orders below.
`conclusion`, 10–231	One point-count threshold, very-general locus, curve multiplicities, volume upper bound	Quantifiers retained; geometric transfers are explicit premises of our final corollaries.

Write $\mathcal{M}(S) = \operatorname{span}_{\mathbb{C}}\{x^a : a \in S\}$ on $(\mathbb{C}^*)^n$. A graded support family satisfies $S_k + S_l \subset S_{k+l}$ for every positive k, l. The source's flag gives

$$S_k \subset k\Delta(1/D, \dots, 1/D, D^{n-1}V), \qquad \#S_k = h^0(X, kL),$$

where DL is very ample and $\Delta(p) = \{x \geq 0 : \sum x_i/p_i \leq 1\}$. The lexicographic valuation is fixed in all degrees; exposing weights need not be. The two-intercept operation replaces A, B by $w, AB/w$ provided $w \max(3/A, 2/B) < 1/16$. Its node normalization uses $d = \max(3/A, 2/B)$, $H = AB$, $W = \max(2A, 3B) = dH$; retaining this last identity is essential to the area calculation. The primitive direction and valuation order are fixed after choosing r, whereas analytic scaling is chosen separately for each degree and tuple. Eventually the source constructs

$$F_k \subset kP \cap \mathbb{Z}^n, \quad P = \Delta(w, \dots, w, rw), \quad \#F_k = h^0(X, kL), \quad F_k + F_l \subset F_{k+l}. \tag{1.1}$$

We distinguish these geometric assertions from all local theorems below.

2 Original Ram inputs and the correct finite map

Fresh local readings of three original private sources informed the normalizations, not the geometric existence claims. TR-rev-1-old-tr250, lines 116–200 and 435–480, defines finite moment coordinates $m = Ec$, explains polynomial unisolvence, and insists that coordinate changes do not create approximation or geometric theorems. Volume 1, Chapter 3, lines 1–210, supplies the Taylor multi-index factorial convention and an actual integral remainder proof. Its output-centered and source-centered formulas differ; neither is silently substituted for algebraic jets here. Its two included auxiliary files were not read. The original `stability.tex`, lines 1–62 and 111–195, proves the finite singular-value perturbation estimate and a nonlinear residual-ball theorem, with a warning about rectangular left inverses. The nonlinear theorem is not needed here.

For holomorphic functions use normalized jets $\partial^\alpha f(q)/\alpha!$, equivalently Taylor coefficients. A jet map $J : \mathbb{C}^N \to \mathbb{C}^D$ must have *full row rank* to prescribe arbitrary data. Applying a full-column-rank

estimate directly to J would be wrong when $N > D$; it applies to J^*. Throughout our finite estimates coefficient and jet coordinates have the stated Euclidean norms, not an unstated geometric L^2 metric. Different frames require their own norm conversion bounds. Classical Vandermonde and CRT facts below are credited as classical; the Ram contribution is a concrete, remainder-controlled realization at the source's particular finite interfaces.

3 Exact Laurent compression and its lifts

Theorem 3.1 (Exact rows)**.** *Let $A = \{a_1 < \cdots < a_q\} \subset \mathbb{Z}$, $q \geq 1$. In $W_A = \operatorname{span}\{z^{a_j}\}$ the possible orders of nonzero germs at $z = 1$ are exactly $0, \ldots, q-1$. Put*

$$B_{\ell j} = \binom{a_j}{\ell}, \quad 0 \leq \ell < q,\ 1 \leq j \leq q, \qquad \binom{t}{\ell} = \frac{t(t-1)\cdots(t-\ell+1)}{\ell!}.$$

Then

$$\det B = \frac{\prod_{i<j}(a_j - a_i)}{\prod_{\ell=0}^{q-1} \ell!} \neq 0. \tag{3.1}$$

For $0 \leq b < q$, the unique lift with its first q Taylor coefficients equal to those of u^b is

$$f_b(z) = \sum_j c_{jb} z^{a_j}, \qquad f_b(1+u) = u^b + O(u^q), \tag{3.2}$$

$$c_{jb} = \sum_{v=0}^{b} (-1)^{b-v} \binom{b}{v} L_j(v), \qquad L_j(t) = \prod_{i \neq j} \frac{t - a_i}{a_j - a_i}. \tag{3.3}$$

These formulas are valid for negative Laurent exponents as well.

Proof. Changing the polynomial basis $1, t, \ldots, t^{q-1}$ to $\binom{t}{0}, \ldots, \binom{t}{q-1}$ multiplies its evaluation determinant by $\prod \ell!^{-1}$. This proves (3.1). The first q Taylor coefficients of $(1+u)^{a_j}$ are its column, including for negative integers, since the binomial expansion converges for $|u| < 1$. Thus every prescribed vector of these coefficients has a unique lift, and no nonzero germ in W_A vanishes to order q.

For every polynomial L of degree below q, Newton's finite-difference identity is $L(t) = \sum_{b<q} (\Delta^b L)(0) \binom{t}{b}$. One can verify it on the binomial basis using $\Delta \binom{t}{b} = \binom{t}{b-1}$. Applying it to L_j and evaluating at the a_i proves that the matrix in (3.3) times B is the identity. It is therefore B^{-1} and gives (3.2). □

This strengthens the source's row enclosure $[0, a_+ - a_-]$ to the *exact* interval $[0, q-1] \cap \mathbb{Z}$. It is an elementary Vandermonde extraction, not a new general interpolation theorem. It avoids row reduction of infinite Taylor expansions and detects unused exponent gaps immediately.

Corollary 3.2 (Combinatorial graded compression)**.** *For each remaining-coordinate label β, let $A_{k,\beta}$ be the nonempty first-coordinate row of S_k. Then*

$$\mathcal{C}_1(S_k) = \bigcup_{\beta} (\{0, \ldots, |A_{k,\beta}| - 1\} \times \{\beta\}).$$

This formula preserves cardinalities and all graded inclusions without any choice of analytic scaling or a compatible system of lifts.

Proof. The exact-row theorem gives the formula. If A, B are finite nonempty integer sets of sizes a, b, their increasing listings give a strictly increasing chain of $a+b-1$ sums: first add the smallest member of B to all members of A, then add each remaining member of B to the largest member of A. Hence $|A+B| \geq a+b-1$. The graded input contains this sumset in the summed row, whose compressed interval therefore contains $[0, a+b-2] \cap \mathbb{Z}$. Empty rows impose no condition. □

Proposition 3.3 (Arithmetic size and analytic remainder)**.** *Put* $H = \max(q, 1, |a_1|, \ldots, |a_q|)$ *and* $D_j = \prod_{i \neq j} |a_j - a_i|$. *Then*

$$|c_{jb}| \leq \frac{2^b (2H)^{q-1}}{D_j}. \tag{3.4}$$

For $0 < \rho < 1$ *define*

$$M_b(\rho) = \sum_j |c_{jb}| \begin{cases} (1+\rho)^{a_j}, & a_j \geq 0, \\ (1-\rho)^{a_j}, & a_j < 0. \end{cases}$$

For real $s > 0$ *and* $T > 0$ *with* $sT < \rho$, *the normalized lift* $g_{b,s}(Z) = s^{-b} f_b(1+sZ)$ *satisfies*

$$\sup_{|Z| \leq T} |g_{b,s}(Z) - Z^b| \leq \frac{M_b(\rho) \rho^{-q} s^{q-b} T^q}{1 - sT/\rho}. \tag{3.5}$$

At $|Z_0| \leq R < T$, *its normalized derivative error of order* j *is at most the right side divided by* $(T-R)^j$.

Proof. For $0 \leq v \leq b < q$, every $|v - a_i| \leq 2H$, so the absolute value of $L_j(v)$ is at most $(2H)^{q-1}/D_j$. Sum (3.3) absolutely. On $|u| = \rho$ the absolute value of $f_b(1+u)$ is at most $M_b(\rho)$. Cauchy's coefficient formula bounds its coefficient of u^ℓ by $M_b(\rho)\rho^{-\ell}$. The first erroneous coefficient has $\ell \geq q$. Sum the geometric series after $u = sZ$ and division by s^b to prove (3.5). Cauchy's formula on a circle of radius $T - R$ centered at Z_0 proves the derivative statement; normalization removes $j!$. □

The exact denominators of c_{jb} divide the signed integer $\prod_{i \neq j}(a_j - a_i)$, because the numerators in (3.3) are integer finite differences of an integer polynomial. The bound is deliberately conservative. Rank does not imply uniformly well-conditioned lifting: for $A = \{0, M\}$ one has $f_1(z) = (z^M - 1)/M$, and the coefficient of u^2 is $(M-1)/2$. Thus fixed nonzero scaling cannot give a uniform small error as $M \to \infty$. For $A = \{-2, 1, 4\}$ the determinant is 27, the compressed row is $\{0, 1, 2\}$, whereas the source's span enclosure extends through 6.

4 A certified finite analytic transfer

Fix a finite output support and the normalized row lifts above, with each column carrying its unchanged Laurent factor U^β. Fix distinct torus test points and their full jets of specified orders, so the limiting matrix J_0 has D rows and N columns. Assume it is surjective. Around each point choose a closed complex polydisc avoiding $U_i = 0$, with coordinate radii $d_{i,h} > 0$, lying in $|Z| \leq T$. Let $E_{ij}(s)$ bound the column-j function error on that polydisc: (3.5) times the supremum of $|U^{\beta_j}|$ is one explicit choice.

Theorem 4.1 (Row-lift transfer certificate)**.** *Index jet rows by* (i, α) *and use normalized Taylor coefficients. Set*

$$\eta(s)^2 = \sum_{i,\alpha,j} E_{ij}(s)^2 \prod_{h=1}^{n} d_{i,h}^{-2\alpha_h}. \tag{4.1}$$

If a certified lower singular-value bound for J_0^* is $\gamma > 0$ and $\eta(s) < \gamma$, then the lifted map J_s is surjective and has a right inverse of norm at most $(\gamma - \eta(s))^{-1}$. The original Laurent space separates the same jets at the image tuple $(Z, U) \mapsto (1 + sZ, U)$, provided $sT < \rho < 1$.

Proof. Multivariable Cauchy estimates bound each entry difference by the corresponding summand's square root in (4.1). Thus $\|J_s - J_0\|_2 \le \|J_s - J_0\|_F \le \eta(s)$. For every jet vector v,

$$\|J_s^* v\|_2 \ge (\gamma - \eta(s))\|v\|_2.$$

Consequently $J_s J_s^*$ is positive definite and $J_s^*(J_s J_s^*)^{-1}$ is the claimed right inverse. The affine substitution has nonzero derivative and is injective, while $1 + sZ \neq 0$ on the discs. It preserves distinctness and induces isomorphisms on truncated local rings. The row-basis change and column normalizations are invertible for $s > 0$. □

This uses the original private perturbation proof in the correct adjoint orientation, with an explicit source-matched analytic remainder. It is not credited as a new singular-value inequality. The parameters $q, H, M_b, \gamma, d_{i,h}$ may all depend on the degree, tuple, and jet order. Applying it to the source's node or flag steps needs bounds for their actual analytic germs; Laurent-row bounds alone do not cover those steps.

5 Finite deficits instead of an unspecified saturation threshold

Let $P \subset \mathbb{R}^n$ be compact and convex with nonempty interior. Put $\Lambda_j = jP \cap \mathbb{Z}^n$, $F_j \subset \Lambda_j$, and $h_j = |\Lambda_j \setminus F_j|$. For this section one only needs the single inclusion $F_l + F_{k-l} \subset F_k$ at the degrees tested.

Theorem 5.1 (Anisotropic finite filling)**.** *Let $1 \le l < k$, $y \in \Lambda_k$, $z = y/k$, and $d_i > 0$ satisfy*

$$z + \prod_{i=1}^{n} [-d_i, d_i] \subset P.$$

With $v = \min(l, k - l)$, if

$$h_l + h_{k-l} < \prod_{i=1}^{n} \lfloor 2v d_i \rfloor, \tag{5.1}$$

then $y \in F_k$. The same condition fills every member of any specified set of y for which the same box radii work.

Proof. Every integer x in the box centered at lz with radii vd_i satisfies $x \in lP$ and $y - x \in (k - l)P$. Each closed interval of length L contains at least $\lfloor L \rfloor$ integers: its count is $\lfloor b \rfloor - \lceil a \rceil + 1$ for endpoints a, b, and the stated lower bound follows. Taking products gives the right side of (5.1). At most h_l choices have $x \notin F_l$ and at most h_{k-l} have $y - x \notin F_{k-l}$, since the second map is injective. A surviving choice and the assumed inclusion prove the assertion. □

One may replace the product by the exact count

$$D_l(y) = \#\big(lP \cap (y - (k - l)P) \cap \mathbb{Z}^n\big), \tag{5.2}$$

or count only holes inside this intersection. This can be substantially stronger than the uniform box. The strict inequality matters: with $P = [0, 1]$, $k = 2, l = 1, y = 1$, take $F_1 = \{0\}$ and $F_2 = \{0\}$. There are two possible decompositions, $h_1 + h_1 = 2$, and 1 need not belong to F_2.

Corollary 5.2 (A degree scale with a declared deficit bound)**.** *Suppose $n \geq 2$, $h_j \leq Cj^{n-1}$ for $j \geq J$, and the box of common radius $\delta > 0$ about each z of a set K lies in P. Let $l = \lfloor k/2 \rfloor$. It suffices that*

$$k \geq 3, \quad l \geq J, \quad l\delta \geq 1, \quad k > 3^n C\delta^{-n}$$

to have $kK \cap \mathbb{Z}^n \subset F_k$.

Proof. For $x \geq 1$, $\lfloor 2x \rfloor \geq x$. Also $l \geq k/3$ and $l^{n-1} + (k-l)^{n-1} \leq k^{n-1}$. The product in (5.1) is at least $(k\delta/3)^n > Ck^{n-1}$. □

This makes the source's two-summand argument a finite test; the source's little-o hypothesis alone supplies no numerical C or J. Anisotropic radii are useful here because the source's last simplex intercept is r times each other intercept. Replacing all radii by their minimum can lose that factor in the sufficient missing-point budget.

6 A minimal staircase for unequal jets

Fix $n \geq 2$, distinct $c_1, \ldots, c_r \in \mathbb{C}^*$, and integers $m_i \geq 0$. For $\beta \in \mathbb{Z}_{\geq 0}^{n-1}$ put

$$\nu_i(\beta) = \max(m_i + 1 - |\beta|, 0), \qquad D_\beta = \sum_i \nu_i(\beta), \qquad E(\mathbf{m}) = \{(\beta, b) : 0 \leq b < D_\beta\}.$$

The finite set omits all β with $D_\beta = 0$. Write $q_i = (1, \ldots, 1, c_i)$.

Theorem 6.1 (Exact jet staircase and determinant)**.** *Evaluation of normalized jets through order m_i at q_i is an isomorphism*

$$\mathcal{M}(E(\mathbf{m})) \longrightarrow \bigoplus_i \mathcal{O}_{(\mathbb{C}^*)^n, q_i}/\mathfrak{m}_{q_i}^{m_i+1}. \tag{6.1}$$

In particular,

$$|E(\mathbf{m})| = \sum_i \binom{n+m_i}{n}. \tag{6.2}$$

With columns ordered by β and then increasing b, in the basis $(x'-1)^\beta Y^b$, and jet rows ordered by β, then i, then increasing Y-order, its determinant is exactly

$$\prod_{i<j} (c_j - c_i)^{T_{ij}}, \qquad T_{ij} = \sum_{\beta \in \mathbb{Z}_{\geq 0}^{n-1}} \nu_i(\beta)\nu_j(\beta). \tag{6.3}$$

The same determinant holds in the monomial basis with the corresponding ordering; arbitrary permutations can of course change its sign.

Proof. There are $\binom{n-2+d}{n-2}$ multi-indices β of size d. Alternatively count directly pairs (β, t) with $|\beta| + t \leq m_i$ to obtain $\sum_\beta \nu_i(\beta) = \binom{n+m_i}{n}$. This proves (6.2).

The set of allowed exponents is downward closed in β for each fixed b: if $\alpha \leq \beta$ componentwise then $D_\alpha \geq D_\beta$. Consequently the centered polynomials $(x'-1)^\beta Y^b$ form a basis of the same monomial space, by a unit triangular change of basis in any linear extension of the componentwise ordering. A normalized x' derivative of order α, evaluated at $x' = 1$, kills every such basis element unless $\alpha = \beta$. The full jet matrix is therefore block diagonal. The β block is the one-variable Taylor evaluation of polynomials of degree below D_β with multiplicities $\nu_i(\beta)$ at c_i.

Here is a determinant proof that also gives surjectivity without an uncomputed rank assumption. For nonnegative integers ν_i, use the monic basis

$$P_i(Y)(Y-c_i)^t, \quad 0 \le t < \nu_i, \qquad P_i(Y) = \prod_{j<i} (Y-c_j)^{\nu_j}.$$

It has consecutively increasing degrees and a unit triangular coefficient matrix relative to $1, Y, \dots, Y^{\sum \nu_i - 1}$. At earlier nodes its required jets vanish. At node c_i its diagonal Taylor block is triangular with diagonal $P_i(c_i)$. Its determinant is $\prod_i P_i(c_i)^{\nu_i} = \prod_{j<i}(c_i - c_j)^{\nu_i \nu_j}$. Zero multiplicities simply omit a block. Taking the product over β gives (6.3), which is nonzero. There are no extra factorials because the derivative coordinates are divided by their factorials. □

For constructive interpolation, solve the one-variable congruences $f_\beta(Y) \equiv h_{i,\beta}(Y - c_i)$ modulo $(Y-c_i)^{\nu_i(\beta)}$, with $\deg f_\beta < D_\beta$, and sum $(x'-1)^\beta f_\beta(Y)$. This is precisely the source's construction, with unequal multiplicities retained. It uses the minimum possible number of independent monomials for those data, not an assertion of a unique optimal support shape. For equal $m_i = m$, it is the source's $E_{r,m}$; the source states surjectivity, while the exact size shows an isomorphism. For $n = 3$ and $(m_1, m_2) = (1, 2)$ the staircase has $4 + 10 = 14$ monomials and determinant $(c_2 - c_1)^{10}$. Equal order 2 at both nodes would instead use 20 monomials. The reduction is useful only when the desired jet orders really are unequal; it is not a gain on the equal-order problem.

7 A finite deficit-to-Seshadri certificate

Return to $P = \Delta(p_1, \dots, p_n)$ with $p_1 = \cdots = p_{n-1} = w$, $p_n = rw$. For the equal-order staircase one has the exact, attained identity

$$\max_{e \in E_{r,m}} \sum_i \frac{e_i}{p_i} = \frac{m + 1 - 1/r}{w} =: S_m. \tag{7.1}$$

Indeed for $|\beta| = d \le m$ the largest allowed last exponent is $r(m+1-d) - 1$. Thus the source's convenient strict bound $m+1$ can be replaced by the exact discrete facet value $m + 1 - 1/r$.

Theorem 7.1 (Finite anisotropic stencil test)**.** *Fix $k \ge 2$, $1 \le l < k$, $t \in \mathbb{Z}^n$, and positive real d_i. Suppose*

$$t_i/k \ge d_i \quad (1 \le i \le n), \qquad \frac{\sum_i t_i/p_i + S_m}{k} + \sum_i d_i/p_i \le 1, \tag{7.2}$$

and (5.1) holds with $v = \min(l, k-l)$. Then $t + E_{r,m} \subset F_k$, provided $F_l + F_{k-l} \subset F_k$ and $F_j \subset jP \cap \mathbb{Z}^n$ at these degrees. Consequently $\mathcal{M}(F_k)$ separates full order-m jets at $(1, \dots, 1, c_i)$ for any distinct nonzero c_i.

If, in addition, the actual backward transfer from $\mathcal{M}(F_k)$ to $H^0(X, kL)$ is established, then $\varepsilon(X, L; \mathbf{p}) \ge m/k$ on a nonempty Zariski-open subset of $U_r(X)$ for $m \ge 1$.

Proof. For each $e \in E_{r,m}$, $z = (t+e)/k$ has $z_i \ge d_i$. Its entire box of radii d_i is in the nonnegative orthant. The largest possible value of the facet functional on that box is at most the left side of the second inequality in (7.2). Thus the box lies in P, and Theorem 5.1 proves the support inclusion. Multiplication by x^t induces an isomorphism on every finite jet algebra at a torus point. Theorem 6.1 proves jet separation.

For completeness, the finite geometric implication uses ordinary curves. Surjectivity on X is open because it is the full-row-rank locus of the map to the locally free principal-parts bundles. At

a tuple in this locus and for an integral curve C through a marked point, choose a nonzero degree-m jet on the curve and lift it to X, with zero lower terms and zero jets at the other points. Such a nonzero curve jet exists by Nakayama's lemma in its one-dimensional local domain. The resulting section of kL does not vanish identically on C and vanishes to order at least m at each marked point of C. If $f \in \mathfrak{m}^m$ is its local equation, then

$$q \operatorname{length}(\mathcal{O}_C/(f)) = \operatorname{length}(\mathcal{O}_C/(f^q)) \geq mq \operatorname{mult}_p C + O(1).$$

Divide by q and sum over the marked points to obtain $kL \cdot C \geq m \sum_i \operatorname{mult}_{p_i} C$. This proves the stated lower bound, including singular curves, without asserting the source's global transfers. □

One can also use $E(\mathbf{m})$: replace S_m by the actual maximum of the same facet functional on its finite staircase. The last curve argument gives $kL \cdot C \geq \sum_i m_i \operatorname{mult}_{p_i} C$ when all $m_i \geq 1$. For prescribed positive weights b_i, the weighted ordinary constant with denominator $\sum_i b_i \operatorname{mult}_{p_i} C$ is then at least $\min_i m_i/(kb_i)$. This is a finite weighted extension, not eventual weighted maximality.

Corollary 7.2 (Quantitative approach at fixed point count)**.** *Assume the full family (1.1) and its backward jet transfers, and suppose a known $C > 0$ satisfies $h_j \leq Cj^{n-1}$ for $j \geq J$. Let $V = \prod_i p_i = rw^n$, $B = \sum_i p_i^{-1}$, $l = \lfloor k/2 \rfloor$, and set*

$$\eta_k = \left(\frac{2\,3^n C}{Vk}\right)^{1/n}, \qquad t_i = \lceil kp_i\eta_k \rceil, \qquad d_i = p_i\eta_k.$$

Whenever $k \geq 3$, $l \geq J$, $l \min_i p_i\eta_k \geq 1$, and

$$m = \lfloor kw(1 - 2n\eta_k) - wB - 1 + 1/r \rfloor \geq 1, \tag{7.3}$$

there is a nonempty open tuple locus with

$$\varepsilon(X, L; \mathbf{p}) \geq \frac{m}{k} \geq w - 2nw\left(\frac{2\,3^n C}{Vk}\right)^{1/n} - \frac{wB + 2 - 1/r}{k}. \tag{7.4}$$

Thus the finite lower certificates approach w at rate $O(k^{-1/n})$ for fixed X, L, r, C, J, conditional on the stated inputs.

Proof. Ceilings give $\sum_i t_i/p_i \leq kn\eta_k + B$. Equation (7.3) implies $S_m \leq k(1 - 2n\eta_k) - B$, proving (7.2). The product in (5.1) is at least

$$l^n V \eta_k^n \geq (k/3)^n V \eta_k^n = 2Ck^{n-1} > h_l + h_{k-l}.$$

Apply Theorem 7.1 and the inequality $\lfloor x \rfloor \geq x - 1$. □

Proposition 7.3 (Obtaining a declared deficit constant)**.** *If $\#F_j = h^0(X, jL)$ and a certified lower Hilbert bound is*

$$h^0(X, jL) \geq \operatorname{vol}(P) j^n - C_H j^{n-1} \quad (j \geq J \geq 1),$$

then one may take

$$C = C_H + \operatorname{vol}(P) nB(1 + B)^{n-1} \tag{7.5}$$

in the preceding corollary, or any larger positive value.

Proof. For each $a \in jP \cap \mathbb{Z}^n$, the unit cube $a + [0, 1]^n$ lies in $(j + B)P$. Their interiors are disjoint, so $|jP \cap \mathbb{Z}^n| \leq \operatorname{vol}(P)(j + B)^n$ even for irrational intercepts. Subtract the lower Hilbert bound and use $(j + B)^n - j^n \leq nB(j + B)^{n-1} \leq nB(1 + B)^{n-1} j^{n-1}$. □

For a concrete finite illustration, take $n = 3$, $r = 2$, $w = 1$, $k = 24$, $l = 12$, $t = (2, 2, 4)$, and $d = (1/12, 1/12, 1/6)$. For $m = 11$, $\sum t_i/p_i = 6$, $S_m = 23/2$, and the facet-box sum is $35/48 + 1/4 = 47/48 < 1$. The box budget is $2 \cdot 2 \cdot 4 = 16$. Thus $h_{12} \leq 7$ and the degree-12 product inclusion already certify $t + E_{2,11} \subset F_{24}$ and full jets of order 11. This is a conditional support example, not an assertion that an arbitrary polarized threefold has that deficit or that backward transfer. A cubic box of the smallest radius would only allow budget 8.

8 What is retained, improved, and not supplied

No substantive false assertion was identified in the locally rederived source interfaces. The following are strengthenings, not corrections of wrong source inequalities: exact row cardinality replaces the weaker span enclosure; normalized lifts retain all q matching Taylor coefficients; the source's qualitative saturation proof becomes a strict finite deficit test; and its equal-order Hermite surjection becomes an exact determinant and an unequal-order staircase. The exact discrete facet improvement is $1/r$, not an asymptotic improvement in the volume bound.

The source correctly retains labels in the other $n - 2$ coordinates, so two-dimensional reshaping does not replace full-dimensional jets by planar ones. It uses both additive supports *in every degree* and analytic rank transfer *at each fixed degree*; these are different obligations. Its single threshold in r is chosen before r is fixed, while k and analytic scales may subsequently depend on r and on the target ratio. Nothing here removes these quantifier distinctions. The surface report M096 uses different final interpolation directions and is not a substitute for this source or its $n \geq 3$ theorem.

The full source route still requires the actual flag, node valuations, primitive lattice maps, bounds for real intercepts, the exact count $\#F_k = h^0(X, kL)$, and the backward geometric transfer. We read their proof text and preserve their role, but do not promote a local certificate to independent certification of the all-r theorem. Bertini, Hilbert polynomial theory, principal parts, and the volume/nef intersection facts are standard imported geometry, not consequences of a Ram transform.

The conditional rate (7.4) has explicit constants only when C_H, J or the deficits themselves have been supplied. It is uniform in the finite stencil for a fixed k, not uniform in all varieties, r, or analytic transfer chains. A countable intersection of the appropriate nonempty open loci over degrees gives a very-general common locus by the Baire argument over $\mathbb{C}$; it neither produces an explicit algebraic tuple nor treats arbitrary special points. Even exact finite rank in many degrees does not establish an all-degree hypothesis. Without additivity, removing one interior lattice point at every degree keeps full asymptotic density but defeats interior saturation.

Benefits and costs. The row formula uses cardinalities alone for support compression; the inverse and Cauchy estimates separately expose its arithmetic cost. Anisotropic boxes exploit the actual elongated geometry, and the staircase uses no excess dimension for its specified data. Conversely the tail bounds may be pessimistic, and Vandermonde factors become poorly conditioned when the chosen nodes cluster. Exact nonzero determinants do not fix that issue. The deficit criterion can fail even when the desired stencil is present; it is sufficient, not necessary. Computation of deficits after the nonlinear source reshaping may be more expensive than a direct finite jet test.

Feasible next work. Use localized hole counts in (5.2) and optimize the split l rather than always halving. For explicit polarized varieties, derive numerical Hilbert remainders and propagate actual analytic norms through the node and flag maps; this would turn the conditional certificates into computed geometric bounds. Compare different torus nodes using the explicit confluent determinant

and certified singular values. None of these tasks is discharged by increasing the size of a finite test suite.

9 Public context and verification

Public-primary searches were made on October 8, 2026, using public titles only. Ito's Proposition 5.7 [2] gives the finite initial-monomial jet comparison, and his Theorem 5.8 places it in graded linear-series theory. Our quantitative row specialization does not claim to originate that degeneration principle. Roé–Ross [3] supplies a product comparison of multipoint constants, not eventual equality for each point count. Trusiani [4] studies multipoint Okounkov bodies and their local positivity and packing information. These primary sources provide context; their complete proofs were not audited here. The fresh search is not an exhaustive priority search and does not independently settle the status of the source's global claim.

The current primary abstracts also report more specialized 2026 results. Ito [5] relates Seshadri constants and Gromov widths to lattice width for generalized-permutohedral toric varieties. Laface–Ugaglia [6] and Malara–Merta–Szpond–Zieliński [7] report irrational surface examples obtained from dihedral configurations, the latter for an infinite odd-indexed family. Pokora–Szemberg [8] report an improved lower bound at ten very general plane points using finite-field lifting. Only their abstracts and metadata were read here. Neither their specialized settings nor this report's finite computations independently establish the source's all-large-r, fixed-variety assertion.

The independent author check script uses exact rational arithmetic for generalized binomial inverses, lift coefficients, sumset inclusion, Hermite jet determinants, and finite lattice-deficit examples. Its accompanying JSON states the actual checked ranges and counting units. Manufactured examples test formulas, not the geometry of a polarized variety. Parent checks, when available, are separate and are not included in author counts. The author run passed 16,772 units: 218 Laurent rows, 14,256 exact rational lift/derivative evaluations, 961 sumset pairs, 60 full multivariate Hermite matrices, 72 facet cases, 1,107 shifted interval counts, 72 deficit models, 24 lattice upper bounds, and two named certificates. The latter include all 728 points of the displayed stencil and an actual row-lift rank test at $s = 1/512$ with $\gamma^2 \geq 1/9$; these detail counts are not added again. The read ledger records unread private complements, omitted companion files, and pre/post hashes. Compilation, all-page visual inspection, and the freeze manifest check are artifact verification only.

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[9] M. Subbarao, original confidential local Ram sources: TR-rev-1-old-tr250, selected moment-coordinate and polynomial proofs; Ram Calculus, Volume 1, Chapter 3, local differential representation; TR1-lin-nonlin, `stability.tex`, finite perturbation proof. Exact local paths, literal intervals, and SHA256 values are recorded in the private read-scope ledger. No private material was sent to public search.